

Course Code: CSE2074	Course Title: Database Management Systems Type of Course: 1) School Core 2) Laboratory Integrated	L-T-P-C	2 - 0	2	3
Version No.	1.0				
Course Pre-requisites	NIL				
Anti-requisites	NIL				
Course Description	This course introduces the core principles and techniques required in the design and implementation of database systems. It covers concepts of relational database systems (RDBMS). More emphasis is set on how to design, develop, organize, maintain and retrieve the information efficiently. It helps the students to learn and practice data modeling and database designs. The associated laboratory is designed to implement database design using MySQL (My Structured Query Language-Open Source) in information technology applications. All the exercises will focus on the fundamentals for creating, populating, sophisticated, interactive way of querying, and simultaneous execution of the transactions of database.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Database Management Systems and attain SKILL DEVELOPMENT through EXPERIENTIAL LEARNING techniques.				
Course Outcomes:	On successful completion of the course the students shall be able to: 1] Understand core concepts of database (Knowledge) 2] Apply normalization techniques to refine database schema (Application) 3] Develop database with concurrent transactions execution feature (Application)				
Course Content					
Module 1	Introduction to Database and its Conceptual Model (Knowledge)	Assignment	Problem Solving	6 Classes	
Topics: Introduction to Database: Schema, Instance, 3-shema architecture, physical and logical data independence, Data isolation problem in traditional file system, advantages of database over traditional file systems. Conceptual Data Modelling: Entity Relationship (ER) Model, ER Model to Relational Model, Examples on ER model.					
Module 2	Query Languages (Application)	Assignment	Problem Solving	7 Classes	

Topics: Relational Algebra with selection, projection, rename, set operations, cartesian product, joins (inner and outer joins), and division operator. Examples on Relational Algebra Operations. MySQL Database Querying, DDL, DML, Constraints, Operators, Set Operators, Aggregate Functions, Joins, Views, Procedures, Functions and Triggers.				
Module 3	Designing and Refining Database Schema (Application)	Assignment	Programming Task	7 Classes
Topics: Schema Design: Problems in schema design, redundancy and anomalies. Schema refinement: Normal Forms based on Primary Keys-(1NF,2NF, 3NF), Boyce-Codd Normal Form, Multi valued Dependency (Fourth Normal Form), Join Dependencies (Fifth Normal Form), lossy and lossless decompositions.				
Module 4	Transaction Management and Concurrency Control (Application)	Assignment	Problem Solving	6 Classes
Topics: Transaction: Desirable properties (ACID) of Transactions, Simultaneous Transactions and their problems like dirty read, lost update and incorrect summary, Serializability, Conflict Serializability, View Serializability; Concurrency Control: Locking and Time-stamping concurrency schemes.				
List of Laboratory Tasks: Create Employee, Student, Banking and Library databases and populate them with required data. Do the following experiments of different lab sheets on those databases. Labsheet-1 [3 Practical Sessions] Experiment No 1: [1 Session] To study and implement Data Definition Language (DDL) commands and Data Manipulation Language (DML) commands of MySQL. Level 1: Perform operations using Data Definition Language and Data Manipulation Language commands including different variants of SELECT on Student DB. Level 2: Identify the given requirements; valid attributes and data types and Perform DDL and DML operations on a given scenario. [Banking Databases] Experiment No. 2: [2 Sessions] To implement different types of MySQL constraints and relational, logical, pattern matching, BETWEEN, IS NULL, IN and NOT IN Special Operators.				

Level 1: Create tables on Banking database using PRIMARY KEY, NOT NULL, UNIQUE, FOREIGN KEY and demonstrate the working of relational, logical, pattern matching, BETWEEN, IS NULL, IN and NOT IN Special Operators on Student Database.

Level 2: Enforce different types of data and referential integrity constraints. Then try queries with special operators based on the student database. [Banking Database].

Labsheet-2 [3 Practical Sessions]

Experiment No. 3: [1 Session]

To try for aggregation of data in to groups and sub-groups using Group by, HAVING clauses and sort data using Order By Clauses.

Level 1: Implement the conjugate of GROUP BY, ORDER BY and aggregate functions on Banking Database.

Level 2: Implement MySQL queries on library database using appropriate clauses and aggregate functions. Also order the data either in ascending and descending order using corresponding clause. [Library databases].

Experiment No. 4: [2 Session]

To study and implement different types of Set and Join Operations [3 Slots]

Level 1: Demonstrate different types of Set Operations (UNION, UNION ALL, INTERSECT, MINUS) and Join Operations (INNER JOINS, OUTER JOINS, CROSS JOIN, NATURAL JOIN) on two or more tables of Banking Database.

Level 2: Use Set and Join operations to retrieve the data from two or more relations(tables) as per the given scenario. [Library databases]

Labsheet-3 [3 Practical Sessions]

Experiment No. 5: [3 sessions]

To study and implement Views, and Procedures in MySQL.

Level 1: Implement MySQL Views, and Procedures in MySQL on Employee database.

Level 2: Analyze the requirement and construct views, and Procedures on Mini Project Domain. [Banking Database]

Labsheet-4 [3 Practical Sessions]

Experiment No. 6: [3 Sessions]

To study and implement Functions, and Triggers in MySQL.

Level 1: Implement MySQL Functions and Triggers in MySQL on Employee database.

Level 2: Analyze the requirement and construct Functions and Triggers on Mini Project Domain. [Banking Database]